

AP ENVIRONMENTAL SCIENCE · THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — every concept tagged for your tier

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

PAGE 1 | MATH OF 3/4/5 · 9 UNITS · EXAM STRUCTURE

About this exam. APES is interdisciplinary — biology + chemistry + earth science + politics. Strong emphasis on **specific case studies** + environmental laws by name + real-world data interpretation. Less math-heavy than other sciences, but you must know specific facts.

THE MATH OF YOUR TARGET

Target	MCQ (80 q, 60%)	FRQ (3 q, 40%)	Approx %	Tier
3 (PASS)	~38 / 80 (~48%)	~14/30 raw	~47-50%	[3] only
4	~48 / 80 (~60%)	~18/30 raw	~58-62%	[3] + [4]
5	~58 / 80 (~73%)	~22-24/30 raw	~70-73%	All

THE 9 UNITS — by exam weight

#	Unit	Weight	Key topics
1	The Living World: Ecosystems	6–8%	Food chains, biogeochemical cycles, biomes
2	The Living World: Biodiversity	6–8%	Ecosystem services, ecological tolerance, succession
3	Populations	10–15%	r/K-selected , life tables, demographic transition, HUMAN POPULATION
4	Earth Systems & Resources	10–15%	Soil , watersheds, atmosphere, plate tectonics
5	Land & Water Use	10–15%	Tragedy of commons, agriculture, fishing, mining, urban land use
6	Energy Resources & Consumption	10–15%	Fossil fuels, nuclear, renewables , energy efficiency
7	Atmospheric Pollution	7–10%	Smog, acid rain, indoor air, noise
8	Aquatic & Terrestrial Pollution	7–10%	Eutrophication, plastic, endocrine disruptors , solid waste
9	Global Change	15–20%	Climate change , ozone depletion, ocean acidification, invasive species

Units 3, 4, 5, 6, and 9 each command 10-20%. Unit 9 (climate change) is the largest single unit. Energy and pollution combined are ~25%.

ENVIRONMENTAL LAWS — must know by name [3]/[4]

Law (year)	What it does
Clean Air Act (1970, amended 1990) [3]	Sets National Ambient Air Quality Standards (NAAQS) for 6 pollutants: CO, NO ₂ , SO ₂ , O ₃ , particulates, lead.
Clean Water Act (1972) [3]	Regulates point-source water pollution. Defines pollutant discharge permits.
Safe Drinking Water Act (1974) [4]	Sets maximum contaminant levels for tap water.
Endangered Species Act (ESA) (1973) [3]	Protects threatened/endangered species + critical habitat.
CERCLA / Superfund (1980) [4]	Cleanup of hazardous waste sites. Polluter pays principle.
RCRA (1976) [4]	Resource Conservation & Recovery Act. Solid + hazardous waste management (cradle-to-grave).
Montreal Protocol (1987) [3]	International. Phased out CFCs to protect ozone layer. Most successful env. treaty ever.
Kyoto Protocol (1997) [4]	First international agreement to limit greenhouse gases. US never ratified.
Paris Agreement (2015) [3]	Updated international climate accord. Voluntary national pledges to limit warming to <2°C.
NEPA (1970) [4]	National Environmental Policy Act. Required Environmental Impact Statements (EIS) for federal projects.

KEY INSIGHT [5] Naming the law in an FRQ answer earns more credit than vague "a law was passed." Memorize the names + years + what they do.

PAGE 2 | MASTER VOCABULARY · the most-tested terms

APES rewards specific terminology. Generic answers earn no points; "the food chain" earns less than "trophic levels with energy lost as heat (~90%) at each transfer."

Term	Definition + tier
Trophic level [3]	Position in food chain. Producers (1st) → 1°/2°/3° consumers. ~10% energy passes per level (90% lost as heat).
Food chain vs food web [3]	Linear (chain) vs interconnected (web) of feeding relationships.
Biogeochemical cycles [3]	Carbon, nitrogen, phosphorus, water cycles. Test loves: N has gaseous reservoir; P does NOT.
Carrying capacity (K) [3]	Max sustainable population in environment. Logistic growth slows as N approaches K.
Exponential vs logistic growth [3]	Exponential (J-curve, unlimited resources, $dN/dt = rN$); logistic (S-curve, slows at K).
r-selected vs K-selected [3]	r: many offspring, no parental care, short life (mosquitos, weeds). K: few offspring, parental care, long life (humans, whales).
Demographic transition [4]	Stages: 1 high birth/death → 2 high birth/lower death (pop boom) → 3 lower birth/death → 4 low both. Industrial nations in 4.
Total Fertility Rate (TFR) [3]	Avg children per woman. Replacement = 2.1. Below = shrinking; above = growing.
Tragedy of the commons [3]	Hardin. Shared resources (atmosphere, fish, grazing) overused without regulation.
Eutrophication [3]	Nutrient pollution → algal bloom → die → bacteria decompose → O ₂ depletion → fish kills.
Bioaccumulation vs biomagnification [4]	Bioaccumulation: toxin builds in ONE organism over time. Biomagnification: toxin concentrates UP the food chain (top predators worst).
Acid rain [3]	pH < 5.6. Caused by SO ₂ + NO _x from coal burning. Damages forests + lakes + buildings.
Ozone depletion vs smog [4]	Stratospheric ozone (good) destroyed by CFCs. Tropospheric ozone (bad) is photochemical smog.
Greenhouse effect [3]	Atmospheric gases (CO ₂ , CH ₄ , H ₂ O, N ₂ O) trap infrared. Natural; enhanced = human-caused warming.
Albedo [3]	Reflectivity. Ice/snow = high albedo. Dark surfaces (oceans) = low. Positive feedback: ice melts → less albedo → more warming.
Endocrine disruptor [4]	Chemicals mimicking hormones. Affect reproduction, development. Examples: BPA, atrazine, PCBs.
LD₅₀ [4]	Dose lethal to 50% of test population. Lower LD ₅₀ = more toxic.
Half-life [4]	Time for 50% of a substance to decay/break down. Critical for radioactive waste + pesticides.
Renewable vs nonrenewable [3]	Renewable: replenished naturally (solar, wind, biomass). Nonrenewable: finite (fossil fuels, uranium). Nuclear is technically nonrenewable.
Hubbert's peak [4]	Predicted oil production peak. Concept: peak then decline.
Sustainable yield [4]	Highest extraction rate that doesn't deplete resource. Key for fisheries, forestry.
Soil horizons [3]	O (organic), A (topsoil), B (subsoil), C (parent material). E sometimes between A and B.
Ecological succession [3]	Primary (bare rock to mature ecosystem); secondary (after disturbance, soil intact). Pioneer species → climax community.
Keystone species [4]	Disproportionate impact on ecosystem. Removal causes cascade. Sea otters, wolves, beavers.
Niche vs habitat [4]	Niche = role/function in ecosystem. Habitat = physical place where it lives.
Ecosystem services [4]	Provisioning (food, water), regulating (pollination, climate), cultural (recreation), supporting (nutrient cycling).

UNITS 1-2 | THE LIVING WORLD: ECOSYSTEMS + BIODIVERSITY

Energy Flow + Trophic Levels [3]

Producers (autotrophs) → Primary consumers (herbivores) → 2° consumers → 3° consumers

~10% RULE: only 10% of energy passes from one trophic level to the next.
90% lost as heat (respiration), waste, undigested.

10x10x10 EXAMPLE: 10,000 kcal of grass → 1,000 kcal in deer → 100 kcal in wolf

IMPLICATION: shorter food chains support more biomass at top.
Apex predators are rare (energy runs out).

Biogeochemical Cycles [3]/[4]

Cycle	Key facts + AP-tested points
Carbon [3]	Photosynthesis ($\text{CO}_2 \rightarrow \text{C}$) + respiration ($\text{C} \rightarrow \text{CO}_2$). Combustion of fossil fuels = MAJOR human input. Oceans absorb ~30% of emissions.
Nitrogen [3]	Atmospheric N_2 (78%) UNUSABLE by most life. Nitrogen FIXATION by bacteria (legume root nodules, lightning) → ammonia. Nitrification: $\text{NH}_3 \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$. Synthetic fertilizers disrupted N cycle dramatically.
Phosphorus [3]	No gaseous phase. Cycles through rock weathering → soil → plants → consumers → decomposers. Often limiting nutrient in ecosystems. Mining for fertilizer.
Water [3]	Evaporation, condensation, precipitation, runoff, infiltration, transpiration.
Sulfur [4]	Mostly underground (rocks). Volcanic emissions natural source. Coal burning main human source → SO_2 → acid rain.

KEY INSIGHT [4] Limiting nutrients matter. **Phosphorus** often limits aquatic ecosystems (no gas reservoir). **Nitrogen** often limits terrestrial. Adding either can cause eutrophication if excessive.

Biomes — characteristics [3]/[4]

Biome	Key features
Tropical rainforest [3]	High temp + rainfall year-round. Highest biodiversity. Poor soil (nutrients in biomass). Equator regions.
Desert [3]	<25 cm/yr precipitation. Hot or cold (Antarctica = cold desert). Low biodiversity. Adaptations: water storage, nocturnal.
Temperate forest [3]	4 seasons. Deciduous trees. Rich soil. Eastern N. America, Europe, East Asia.
Grassland [3]	Moderate rainfall, prevents trees. Fire-adapted. Prairie, savanna. Most converted to agriculture.
Tundra [3]	Cold, short summers. Permafrost. Low biodiversity. Major carbon sink (frozen soil).
Boreal/Taiga [4]	Cold, snowy. Coniferous trees. Largest terrestrial biome. Important for global oxygen.

Ecological Succession [3]

- **Primary succession** — starts on BARE ROCK (no soil). After volcanic eruption, glacier retreat. Lichens are pioneer species. **Slow** — soil must form first.
- **Secondary succession** — after disturbance leaves soil intact (fire, abandoned farm). **Faster** because soil + seeds remain.
- **Climax community** — relatively stable mature community. Idealized — real ecosystems are dynamic.

Biodiversity + Ecosystem Services [4]

- **Genetic, species, and ecosystem diversity.** Higher diversity = more resilient ecosystems.
- **Ecosystem services:** provisioning (food, fiber, water), regulating (climate, pollination, water filtration), cultural (recreation, spiritual), supporting (nutrient cycling, soil formation).
- **Threats to biodiversity (HIPPCO):** Habitat destruction, Invasive species, Pollution, Population (human), Climate change, Overexploitation.

UNITS 3-4 | POPULATIONS + EARTH SYSTEMS & RESOURCES

Population Growth [3]/[4]

EXPONENTIAL (J-curve): $dN/dt = rN$ (unlimited resources)
 LOGISTIC (S-curve): $dN/dt = rN(K-N)/K$ (K = carrying capacity)

Population growth rate (humans):

$$r = (\text{births} - \text{deaths} + \text{immigration} - \text{emigration}) / \text{total population}$$

$$\text{Doubling time} \approx 70 / r\% \quad (\text{Rule of 70})$$

Total Fertility Rate (TFR): replacement = 2.1.

TFR > 2.1 → growing population

TFR < 2.1 → shrinking (or aging)

Demographic Transition Model [3]/[4]

STAGE 1: Pre-industrial – high birth + high death rates → low growth
 STAGE 2: Transitional – high birth, FALLING death rate → POPULATION BOOM
 STAGE 3: Industrial – falling birth + low death → slowing growth
 STAGE 4: Post-industrial – low birth + low death → stable / shrinking
 STAGE 5 [4]: declining (some Europe + Japan) – birth < death → DECLINING

Most developed nations: stage 4 or 5.

Most developing nations: stage 2-3.

KEY INSIGHT [5] AP loves population pyramid interpretation. Wide base = young/growing population (sub-Saharan Africa). Even/narrowing base = stable/declining (Japan, Germany). Pyramids predict future labor force, dependency ratios, services demand.

Earth's Atmosphere [3]/[4]

Layers (bottom to top):

TROPOSPHERE (0-12 km): weather, where we live, where pollution accumulates

STRATOSPHERE (12-50 km): contains OZONE LAYER

MESOSPHERE (50-85 km): meteors burn up

THERMOSPHERE (85-600 km): aurora borealis, very thin air

Composition: 78% N₂, 21% O₂, 1% Ar, 0.04% CO₂, trace others

Soil Composition + Horizons [3]/[4]

- **O horizon [3]**: organic matter (litter, humus). Top layer.
- **A horizon [3]**: topsoil. Mineral + organic mix. Most biological activity.
- **E horizon [4]**: leached zone (sometimes between A and B).
- **B horizon [3]**: subsoil. Mineral accumulation from leaching above.
- **C horizon [3]**: parent material. Weathered rock.

Soil texture = relative % sand, silt, clay. Loam (balanced) is best for agriculture. **Clay** holds water but compacts. **Sand** drains fast.

Plate Tectonics [3]

- **Convergent boundary** — plates collide. Mountains (Himalaya), volcanoes (Andes), trenches (Mariana).
- **Divergent boundary** — plates separate. Mid-ocean ridges, rift valleys.
- **Transform boundary** — plates slide past. Earthquakes (San Andreas).

Watersheds [3]/[4]

Watershed = land area that drains to a particular river/lake. Pollution upstream affects everything downstream. Protected by **riparian buffers** (vegetation along banks).

UNIT 5 | LAND & WATER USE · agriculture, fishing, mining, urban

Tragedy of the Commons [3]

Garrett Hardin (1968). Shared resources (atmosphere, ocean fisheries, grazing land) get overused because individual users gain while costs are shared.

Solutions: regulation, privatization, community management (Ostrom).

Agriculture — methods + impacts [3]/[4]

Method	Description + impacts
Slash-and-burn [3]	Cut + burn forest. Short-term fertile (ash) but quickly depleted. Drives tropical deforestation.
Monoculture [3]	Single crop. Efficient but vulnerable to pests + disease. Reduces biodiversity. Heavy fertilizer + pesticide use.
No-till farming [4]	Don't plow. Reduces soil erosion + retains moisture + sequesters carbon.
Crop rotation [3]	Alternate crops by year. Replenishes nitrogen (legumes), breaks pest cycles.
Concentrated Animal Feeding Operation (CAFO) [4]	Industrial livestock. Cheap meat, but huge waste/water/air pollution + antibiotic resistance.
Green Revolution [4]	1940s-70s. High-yield seeds + fertilizers + irrigation. Massive food increases but environmental costs (pesticide runoff, soil degradation).
GMOs [4]	Genetically modified crops (Bt corn, Roundup-ready soy). Reduce pesticide use sometimes; concerns about gene flow + resistance.
Sustainable / organic [4]	Lower yields, but reduced inputs, healthier soil, less environmental impact.

Soil Degradation [3]/[4]

- **Erosion** — wind/water removing topsoil. **Dust Bowl (1930s)** = classic case. Solutions: cover crops, contour plowing, terracing.
- **Salinization** [4] — irrigated soils accumulate salts as water evaporates. Renders soil unfit for agriculture. Common in arid regions.
- **Desertification** [4] — productive land becomes desert from overgrazing, deforestation, drought. Sahel region of Africa = textbook example.
- **Compaction** — heavy machinery compresses soil; reduces water infiltration.

Fisheries [3]/[4]

- **Overfishing** [3] — extracting fish faster than they reproduce. ~1/3 of global fisheries overfished.
- **Bycatch** [4] — non-target species caught + killed. Shrimp trawlers especially bad (~5 lbs bycatch per lb shrimp).
- **Bottom trawling** [4] — drags nets along seafloor. Destroys habitat.
- **Aquaculture** [4] — fish farming. Reduces wild pressure but pollution + escapes can harm wild populations.

Mining [3]/[4]

- **Surface mining** [3] — strip mining, mountaintop removal. Cheap but devastating to landscape.
- **Subsurface (deep) mining** [3] — tunnels. Less surface damage but dangerous, expensive.
- **Acid mine drainage** [4] — pyrite + water + O₂ → sulfuric acid → contaminates rivers.

Water Resources + Conservation [3]/[4]

- **Aquifers** — underground water reservoirs. **Ogallala** (US Great Plains) — depletion threatens agriculture.
- **Salinization of irrigated soils** + groundwater depletion are major problems.
- **Drip irrigation** [4] — most efficient (90%+ water reaches plants). Vs flood irrigation (~50%).
- **Dams** trade-offs: flood control + hydropower + reservoir BUT block fish migration + sediment accumulation + displaced communities + downstream ecosystem damage.

UNIT 6 | ENERGY RESOURCES & CONSUMPTION

Fossil Fuels — the big 3 [3]/[4]

Fuel	Pros / cons
Coal [3]	Cheapest, abundant. DIRTIEST — high CO ₂ , SO ₂ , particulates, mercury, ash. Mountaintop removal mining. Acid rain source.
Oil [3]	Energy-dense, transportable. Used for transport. CO ₂ + NO _x emissions. Spills (Exxon Valdez 1989, Deepwater Horizon 2010). Peak oil debate.
Natural gas [3]	Cleanest fossil fuel (~half CO ₂ of coal). Methane leaks during extraction = potent GHG. Fracking concerns: water contamination, induced earthquakes.

Nuclear Energy [3]/[4]

- **Fission** — uranium-235 splits → energy + neutrons → chain reaction in reactor.
- **Pros [3]**: no CO₂ during operation, dense energy, reliable baseload.
- **Cons [3]**: radioactive waste (long half-lives), risk of meltdown, expensive plants, weapons proliferation.
- **Major incidents [3]**: Three Mile Island (1979 US), Chernobyl (1986 USSR), Fukushima (2011 Japan).
- **[5] Yucca Mountain** — proposed US permanent waste repository. Politically blocked. Currently waste stored on-site at reactors.

Renewable Energy [3]/[4]

Source	How it works	Pros / cons
Solar (PV) [3]	Photovoltaic cells convert sunlight to electricity	No emissions; intermittent (need storage); manufacturing has impacts
Solar thermal [4]	Mirrors concentrate sun to heat fluid → steam → turbine	Reliable in sunny areas; high cost; needs lots of land
Wind [3]	Turbines convert wind to electricity	Clean; intermittent; bird/bat kills; aesthetic objections
Hydroelectric [3]	Dam stores water; falling water spins turbine	Renewable, reliable; ecological impacts of dam
Geothermal [4]	Heat from Earth's interior. Iceland leader.	Clean; site-specific; H ₂ S emissions possible
Biomass / biofuels [4]	Burning organic material. Ethanol from corn, biodiesel.	Renewable but burning produces CO ₂ ; can compete with food crops
Tidal / wave [5]	Tidal flow / wave motion → turbine	Predictable; site-specific; ecological impacts
Hydrogen fuel cell [4]	H ₂ + O ₂ → water + electricity	Clean at point of use; H ₂ production usually fossil-fuel-based

Energy Conservation + Efficiency [3]/[4]

- **Efficiency** = useful output / total input. Coal plants ~33%, modern gas turbines ~60%.
- **LED lights** ~80% efficient vs incandescent ~5%. Massive savings opportunity.
- **Cogeneration / CHP** [4] — combined heat + power. Captures "waste" heat from electricity generation. >80% efficient.
- **Negawatts** [5] — energy NOT used through efficiency. Cheaper than building new generation.

KEY INSIGHT [4] **Embodied energy** = total energy required to produce + transport + dispose of a product. Often dwarfs operational energy. Example: producing aluminum requires huge electricity (recycling = 95% energy savings).

UNITS 7-8 | POLLUTION · air, water, land, indoor

Atmospheric Pollution [3]/[4]

Pollutant	Sources + effects
CO [3]	Incomplete combustion. Binds hemoglobin → suffocation.
SO ₂ [3]	Coal burning. Causes ACID RAIN. Respiratory problems.
NO _x [3]	Vehicles + power plants. Acid rain + photochemical smog precursor.
Particulate matter (PM _{2.5} , PM ₁₀) [3]	Soot, dust. Lung damage. Especially harmful in cities.
Tropospheric ozone (O ₃) [4]	Bad in lower atmosphere. Photochemical smog (NO _x + VOCs + sunlight). Damages plants + lungs.
VOCs [4]	Volatile Organic Compounds. From paints, gasoline, plants. Smog precursor.
Lead (Pb) [4]	Phased out from gasoline (US 1986). Still in old paint, batteries. Neurological damage.

Smog — 2 types [3]/[4]

- **Industrial smog (gray-air, sulfurous)** — from coal burning. SO₂ + particulates. Common in older industrial cities. London 1952 "Great Smog" killed ~12,000.
- **Photochemical smog (brown-air)** — NO_x + VOCs + sunlight → ozone + PAN. Common in sunny cities like LA, Mexico City, Beijing.
- **Thermal inversion** [4] — warm air traps cool polluted air below. Worst in valleys (Mexico City, LA).

Acid Rain [3]/[4]

Normal rain pH ~5.6 (CO₂ in atmosphere). **Acid rain pH < 5.6** (often 4.0-4.5). Caused by SO₂ + NO_x reacting with water → H₂SO₄ + HNO₃.

- **Effects:** acidifies lakes (kills fish at pH < 5), damages forests (nutrient leaching), erodes buildings + monuments.
- **[3] US Clean Air Act amendments (1990)** introduced cap-and-trade for SO₂. Successful — emissions down >50%.

Indoor Air Pollution [3]/[4]

- **Radon** [3] — naturally occurring radioactive gas from rock. Seeps into basements. **#2 cause of lung cancer** in US.
- **Asbestos** [3] — fibrous mineral once used as insulation. Causes mesothelioma when fibers inhaled.
- **Formaldehyde** [4] — in particle board, foam insulation. Carcinogen + respiratory irritant.
- **Lead paint** [3] — pre-1978 homes. Especially dangerous to children.
- **Carbon monoxide** [3] — faulty furnaces, attached garages. Silent killer.
- **Tobacco smoke** [3] — leading cause of indoor pollution + premature death.

Water Pollution — major types [3]/[4]

- **Eutrophication** [3] — N + P enrichment from agriculture/sewage → algal bloom → die → bacterial decomposition consumes O₂ → fish kill ("dead zones" — Gulf of Mexico, Chesapeake Bay).
- **Pathogens** [3] — sewage contamination. Cholera, E. coli, hepatitis. Most common pollution problem in developing nations.
- **Heavy metals** [4] — Hg (mercury) bioaccumulates in fish; Pb in old pipes; Cd, As. **Minamata disease (Japan, 1950s)** = mercury poisoning case study.
- **Plastic pollution** [3] — Great Pacific Garbage Patch. Microplastics in food chain.
- **Endocrine disruptors** [4] — BPA, PCBs, atrazine. Mimic hormones. Affect fish reproduction; suspected human effects.
- **Thermal pollution** [4] — heated water from power plants reduces dissolved oxygen, kills fish.

Solid Waste [3]/[4]

- **Sanitary landfills** [3] — sealed with liners. Methane production (can be captured for energy).
- **Incineration** [4] — reduces volume + can produce energy. Releases CO₂, dioxins; ash is toxic.
- **E-waste** [4] — fastest-growing waste stream. Heavy metals, often shipped to developing nations.
- **Reduce > Reuse > Recycle** hierarchy.

UNIT 9 | GLOBAL CHANGE · climate, ozone, ocean, invasives

Climate Change — the basics [3]

Greenhouse gases (GHGs) trap infrared radiation in atmosphere. Without them, Earth would be ~33°C colder. **Enhanced greenhouse effect** = human-caused increase in GHG concentrations → warming.

GREENHOUSE GASES (by warming potential):

- CO₂ — most abundant, longest-lived. Fossil fuel burning + deforestation.
- Methane — 25× CO₂ per molecule. Cattle, rice paddies, landfills, gas leaks.
- N₂O — ~300× CO₂. Fertilizers, cars.
- CFCs/HFCs — extremely potent. Phased out under Montreal Protocol.
- Water vapor — feedback loop, not directly emitted

CO₂ levels: Pre-industrial ~280 ppm. Today ~420 ppm (rising).

Climate Change — Impacts [3]/[4]

- [3] **Sea level rise** — thermal expansion + ice melt. Threatens coastal cities, low-lying nations.
- [3] **Extreme weather** — more intense hurricanes, droughts, heatwaves, wildfires.
- [3] **Ecosystem disruption** — species range shifts, coral bleaching, phenological mismatches.
- [4] **Ocean acidification** — CO₂ + H₂O → carbonic acid → lower pH → calcium carbonate dissolves → coral + shellfish damaged.
- [4] **Permafrost melt** — releases more methane → positive feedback.
- [4] **Albedo feedback** — ice/snow melt → less reflectivity → more warming → more melt.

Ozone Depletion [3]/[4]

Stratospheric ozone (O₃) absorbs UV radiation. CFCs (chlorofluorocarbons) — used in refrigerants, aerosols — broke down ozone. "Ozone hole" formed over Antarctica.

- [3] **Effects of UV increase**: skin cancer, cataracts, immune suppression, plant damage.
- [3] **Montreal Protocol (1987)** — phased out CFCs globally. **Most successful environmental treaty ever** — ozone is recovering.

WATCH OUT Don't confuse ozone depletion with climate change. Different problems, different gases. Ozone depletion = stratospheric ozone destroyed by CFCs. Climate change = greenhouse warming from CO₂, methane, etc.

Invasive Species [3]/[4]

- **Definition**: non-native species that cause ecological/economic harm.
- **Why successful**: lack natural predators in new ecosystem; outcompete natives.
- **Famous examples [4]**: Zebra mussels (Great Lakes), Asian carp (Mississippi), kudzu (US Southeast), brown tree snake (Guam — wiped out native birds), cane toads (Australia), European starlings (US).
- **Pathways**: ballast water, intentional introduction, accidental escape, pet trade.

Biodiversity Loss [3]/[4]

- **6th mass extinction** — current rate of extinction 100-1000× background rate.
- **HIPPCO drivers**: Habitat destruction, Invasive species, Pollution, Population, Climate change, Overexploitation.
- **Conservation strategies**: protected areas, habitat corridors, captive breeding, ESA, **CITES** (international wildlife trade convention).

PAGE 9 | FAMOUS CASE STUDIES · the names AP loves

APES rewards specific case studies. Knowing names + years + what happened earns points where vague references earn zero.

Case study (year)	What happened + significance
Love Canal (1978)	Chemical waste buried under NY neighborhood. Health problems revealed. Catalyzed CERCLA/Superfund.
Three Mile Island (1979)	Partial meltdown at PA nuclear plant. No deaths but eroded US public support for nuclear.
Bhopal disaster (1984)	Methyl isocyanate gas leak at Union Carbide plant in India. ~3,800+ deaths. Worst industrial accident.
Chernobyl (1986)	Reactor #4 explosion in Ukraine (USSR). Massive radiation release. ~30 immediate deaths, thousands of long-term cancers. Exclusion zone still uninhabitable.
Exxon Valdez (1989)	Tanker spilled 11M gallons of oil in Prince William Sound, AK. Decades of ecological damage.
Ozone hole over Antarctica (1985 discovery)	Identified by British scientists. Led to Montreal Protocol.
Dust Bowl (1930s)	Massive soil erosion in US Plains from poor farming + drought. Caused by overplowing grasslands.
Aral Sea shrinking (1960s+)	Soviet irrigation diverted feeder rivers to grow cotton. Sea shrunk to ~10% original size by 2000s. Worst environmental disaster of 20th century.
Cuyahoga River fires (1969+ multiple)	Polluted Cleveland river caught fire. Catalyzed Clean Water Act.
Silent Spring (Rachel Carson, 1962)	Book exposed DDT impacts (especially bird eggshell thinning). Launched modern environmental movement.
Easter Island collapse (~1500s)	Polynesian society on isolated island deforested itself, leading to societal collapse. Cautionary tale of unsustainable resource use.
Deepwater Horizon (2010)	BP oil rig explosion in Gulf of Mexico. Largest marine oil spill in US history. Long-term ecosystem damage.
Flint water crisis (2014+)	Switching to corrosive river water without proper treatment leached lead from old pipes. Environmental justice case — disproportionately affected poor + minority residents.
Fukushima Daiichi (2011)	Japanese nuclear plant damaged by tsunami. Meltdown. Reignited global nuclear safety debate. Germany shut down all nuclear plants.
Great Pacific Garbage Patch	Plastic waste accumulation in North Pacific Gyre. Symbol of global plastic pollution crisis.
Beijing air pollution (2010s)	Severe smog days closed schools, grounded flights. Driven by coal burning + cars. Now improving with regulations.

KEY INSIGHT [5] FRQ trick: when an FRQ describes a generic environmental scenario, check if any of these case studies match — citing the actual case earns context credit.

PAGE 10 | FRQ STRATEGY + 3/4/5 SCORE LADDER

APES FRQ Structure [3]

3 FRQs total — 70 minutes. Each scored on 10-point scale. The 3 are typically:

- **FRQ 1: Design an Investigation** — given a scenario, design an experiment. Identify hypothesis, IV, DV, controls.
- **FRQ 2: Analyze an Environmental Problem + Propose Solutions** — given a problem (e.g., a polluted lake), describe causes + propose mitigation.
- **FRQ 3: Analyze an Environmental Problem with Quantitative Data** — calculations + interpretation. Includes math!

UNIVERSAL FRQ DISCIPLINE [3]/[4]/[5]

- **[3] Use specific terminology** — not casual language. "Eutrophication" not "algae problem."
- **[3] Cite specific case studies + laws** when relevant.
- **[4] Show all math** with units. APES math is calculator-allowed but show your work.
- **[4] Discuss BOTH causes + effects + solutions** when prompt asks about a problem.
- **[5] Connect to multiple units** — a question about smog connects to atmospheric chemistry + transportation + health + economic incentives.

THE SCORE LADDER

3-scorer writes...	4-scorer writes...	5-scorer writes...
"Pollution caused fish to die."	"Eutrophication from agricultural runoff caused algal blooms; bacteria decomposing dead algae depleted O ₂ , killing fish."	"Excess nitrogen + phosphorus from synthetic fertilizers caused eutrophication : algal blooms shaded out submerged plants, then died and bacterial decomposition created hypoxic dead zones with O ₂ < 2 mg/L. Similar to the Gulf of Mexico dead zone caused by Mississippi River runoff. Solutions: reduce fertilizer use, riparian buffer strips, nutrient management plans under the Clean Water Act. "
"Use solar panels."	"Switching from coal to solar reduces CO ₂ emissions and acid rain precursors."	"Replacing coal with solar PV reduces CO ₂ (climate change), SO ₂ (acid rain), and particulates (respiratory disease). Trade-offs: solar manufacturing has its own environmental costs (rare earth mining); intermittency requires battery storage. The 1990 Clean Air Act amendments' SO₂ cap-and-trade program demonstrates that market mechanisms can drive emission reductions effectively."
Names a generic example	Names a specific case study	Names case study + year + significance + connects to environmental law that resulted
States effect	States cause + effect	States cause + mechanism + effects (multiple) + interventions + trade-offs

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — **bag ALL of these:**

- ✓ MCQ: 38+ correct. Don't leave anything blank.
- ✓ Memorize the major laws (CAA, CWA, ESA, Montreal, Paris, NEPA, RCRA).
- ✓ Memorize the biogeochemical cycles + 10% energy rule.
- ✓ Each FRQ: attempt every part with relevant terminology.

For a 4 — **add these on top:**

- ✓ MCQ: 48+ correct.
- ✓ Cite at least 2-3 specific case studies in FRQ answers.
- ✓ Math: practice unit conversions, half-life, percent change.
- ✓ Distinguish ozone depletion from climate change clearly.
- ✓ Know HIPPOCO drivers + DTM stages + r/K-selected differences.

For a 5 — **add these on top:**

- ✓ MCQ: 58+ correct.
- ✓ FRQs: connect across multiple units (e.g., agriculture → soil → water → eutrophication → atmosphere).
- ✓ Discuss trade-offs of every solution proposed.
- ✓ Cite case studies WITH years + locations + significance.
- ✓ Quantitative interpretation: state magnitudes, rates, comparisons.
- ✓ Address environmental justice when relevant (Flint, Cancer Alley, climate refugees).

THE ONE THING THAT MATTERS MOST [3][4][5]: SPECIFIC > GENERIC. APES rewards specific facts — laws by name, case studies by year, mechanisms by step. "Pollution is bad" earns 0; "The Clean Water Act of 1972 sets effluent limits via NPDES permits, helping reduce point-source water pollution like the Cuyahoga River fires that helped catalyze it" earns full credit.

If you do nothing else: master the laws, the biogeochemical cycles, and 10+ specific case studies. Specificity is the entire game on APES.

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